

Image Recovery for Low Earth Orbit by Leveraging Turbulence and Light Curves

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This paper shows a novel method to characterize human-made objects in low Earth orbit (LEO) using compressed sensing on light curve measurements. The proposed approach minimizes total variation to recover a resolved object image from a fully unresolved light curve and a so-called point spread function (PSF) map. The light curves are generated through numerical wave propagation, which considers atmospheric turbulence under anisoplanatic conditions. Subsequently, the light curve model is transformed into a linear measurement model to apply compressed sensing techniques. Notably, the sensing matrix is found to be a superposition of spatially variable PSFs, which significantly downsamples the ideal object image. The proposed approach robustly recovers clear images of objects in LEO, even with imperfect PSF map estimates and Poisson noise in the light curve measurement.

Nomenclature

a	=	circular aperture mask of telescope
C_n^2	=	refractive index structure function, $m^{-2/3}$
$C_{n_i}^2$	=	refractive index structure function of i th phase screen, $m^{-2/3}$
$c_{n,m}$	=	Fourier coefficients of phase screen, rad
D	=	diameter of aperture mask, m
$\mathcal{F}, \mathcal{F}^{-1}$	=	Fourier transform operator and its inverse
f_x, f_y	=	X and Y components of spatial frequency, m^{-1}
g	=	gain of sensor, e^{-1} /analog-to-digital unit (ADU)
H	=	optical transfer function of composite of free space and turbulence
H_{atm}	=	optical transfer function of turbulence
H_{dif}	=	optical transfer function of free space
h	=	point spread function
$h_{\Delta z_i}$	=	impulse response function of the free space
I_{sun}	=	solar intensity at 1 astronomical unit, Wm^{-2}
j	=	imaginary unit
K	=	number of nonzero pixels in sparse coefficient vector
k	=	wave number
L	=	total propagation distance, km
$L_{\text{free}}, L_{\text{turb}}$	=	propagation distance through free space and turbulence, km
ℓ	=	effective focal length of telescope, m
ℓ_0, L_0	=	inner and outer scales, m
m	=	length of compressed image vector, pixels
mag_{sun}	=	visual magnitude of sun
N	=	side length of local phase screen, pixels
$\mathcal{N}$	=	number of extended phase screens
N_{ext}	=	side length of extended phase screen, pixels
n	=	length of image vector, pixels
r_0	=	coherence diameter, m
r_{0_i}	=	coherence diameter of i th phase screen, m
t	=	time, s

u	=	complex amplitude of optical wave, $\sqrt{Wm^{-2}}$
X, x	=	true object image in form of a two-dimensional array and column vector, respectively
x, y	=	X and Y coordinates of point in optical wave, m
y	=	compressed image in form of a column vector
y_j	=	j th entry of compressed image
Z, z	=	turbulence-degraded image in form of two-dimensional array and column vector, respectively
$\tilde{Z}$	=	turbulence-degraded image with Poisson noise
α	=	switch parameter for atmospheric optical transfer function
γ	=	sparse coefficient vector of image
Δz	=	propagation step in turbulence simulation, km
δ	=	Nyquist grid spacing of phase screen, m
δ_f	=	Nyquist grid spacing on focal plane, m
δ_{PSF}	=	Nyquist grid spacing of point spread function, m
δ_{sensor}	=	pixel pitch of sensor, m
θ_0	=	isoplanatic angle, μrad
λ	=	wavelength, m
μ, ν	=	X and Y coordinates on focal plane, m
ρ	=	radial frequency coordinates, m^{-1}
ρ_c	=	cutoff frequency
σ_x^2	=	log-amplitude variance
Φ	=	sensing matrix
Φ_j	=	square matrix obtained by reshaping j th row of sensing matrix
ϕ_j^T	=	row of sensing matrix
ϕ, ϕ_H, ϕ_L	=	phase screen and its high and low-frequency components, rad
Ψ	=	dictionary

I. Introduction

SPACE surveillance is becoming increasingly important as the space environment becomes more crowded. The goal of space surveillance is to detect, track, and identify space objects to predict their orbits and to support future space missions, such as active debris removal. Ground-based observations can provide astrometric observations allowing for orbit determination. However, because of non-conservative orbital perturbation forces, more accurate long-term orbit prediction requires a deeper understanding of the object's characteristics, such as its shape, rotation, and surface properties. One method for characterizing such properties is to analyze a light curve, a time series of the brightness measurements, which gives information beyond the center of mass.

Early studies of light curve inversion [1,2] have focused on the shape and attitude inversion of asteroids. Most asteroids have a convex shape and uniform reflectivity, allowing researchers to recover

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the Extended Gaussian Image (EGI) from the light curve and invert the shape by solving the Minkowski problem. In addition, many asteroids have a single rotation axis, simplifying the attitude inversion problem.

In contrast, light curve inversion for artificial space objects presents a more challenging problem. Many assumptions about asteroids do not hold for these objects; they often have nonconvex shapes and nonuniform reflectivity, resulting in nonunique and high-dimensional solutions. Many studies have addressed this issue by making assumptions about the attitude or albedo shape to reduce the number of unknown states. Yanagisawa and Kurosaki [3] estimated the shape and motion of a Cosmos rocket body by fitting an ellipsoid model. Linares et al. [4,5] used filtering methods for shape and attitude inversion while assuming a simple spin or simple shape. Linares and Crassidis [6] solved shape and attitude inversion using a Markov chain model while assuming a simple shape. Fan and Frueh [7,8] developed a multihypothesis scheme to rank candidate solutions for shapes. Robinson and Frueh [9] augmented EGI-based methods by embedding a concavity creation algorithm. In recent years, several studies have used deep-learning techniques for shape characterization [10–15]. Still, most of these studies do not explicitly recover the object shape, instead classifying it into several predefined classes. All of these approaches reconstructed a three-dimensional state and are plagued by the inherent ambiguity of the system and the curse of dimensionality. As an alternative, this study explores the option of reconstructing a two-dimensional image with a frozen attitude state using compressed sensing.

Compressed sensing (CS) [16–18] is a signal compression technique that significantly reduces sampling rates well below the limit of the Nyquist–Shannon theorem [19]. A high-dimensional signal is mapped to a lower-dimensional signal using a sensing matrix. The sensing matrix must satisfy the Restricted Isometry Property (RIP) and is typically designed with random entries. If the signal is sparse in a particular basis, the signal can be reconstructed by seeking the sparsest solution that explains the compressed signal.

Previous studies investigated the use of compressed sensing in reconstructing the two-dimensional geometry of an unknown space object. A preliminary study [20] discovered and provided the mathematical framework to exploit a similarity between CS-based image compression and light curve measurement under the influence of atmospheric turbulence. An image of an unknown space object was recovered with a simplified light curve model. A subsequent study [21] developed a more comprehensive physics-based model of the light curve. The atmospheric degradation of the light curve is modeled using point spread functions (PSFs) of the atmospheric turbulence. The modified light curve model resulted in a poor RIP condition of the sensing matrix, necessitating the modification of the recovery algorithm.

This study investigates the impact of noise in the light curve and PSF on image reconstruction. The PSF is degraded by adding white noise to model the estimation error. The light curve is degraded by Poisson noise to model the shot noise inherent to optical measurements. The combination of Poisson noise in the light curve and Gaussian noise in the PSF map makes image recovery more challenging. However, this study uses total variation (TV) minimization with a Dantzig selector constraint for robust image reconstruction. The image recovery performance is tested for various object brightness levels and Gaussian noise levels in the PSF map.

II. Compressed Sensing

A. Theory

The mathematical framework of CS is shown here in abbreviated form as relevant to the methods employed in this paper. CS works by jointly compressing and sampling a high-dimensional image $\mathbf{x} \in \mathbb{R}^n$ through a linear operation given by [16]

$$\mathbf{y} = \Phi \mathbf{x} \quad (1)$$

where $\mathbf{y} \in \mathbb{R}^m$ ($m \ll n$) is a compressed signal vector and $\Phi \in \mathbb{R}^{m \times n}$ is a so-called sensing matrix. Because the number of measurements is

significantly smaller than the number of pixels in the image, the system in Eq. (1) is underdetermined.

However, the signal can be accurately recovered by considering the sparsity of the image of interest. Most signals possess a sparse representation, given by $\mathbf{x} = \Psi \boldsymbol{\gamma}$, where a coefficient vector $\boldsymbol{\gamma}$ has only K ($\ll n$) nonzero pixels. The matrix $\Psi \in \mathbb{R}^{n \times n}$ is called a dictionary, and its columns work as bases of the image $\mathbf{x}$. Popular choices for the dictionary in image compression include a discrete cosine transform matrix, a Gaussian matrix, and a Fourier transform matrix. This sparse representation provides an alternative formulation for the compressed image, as given by $\mathbf{y} = \Phi \Psi \boldsymbol{\gamma}$, and the sparse vector is recovered by seeking the sparsest solution that explains the compressed image [16–18], as given by

$$\hat{\boldsymbol{\gamma}} = \arg \min_{\boldsymbol{\gamma}} \|\boldsymbol{\gamma}\|_1 \quad \text{subject to} \quad \|\mathbf{y} - \Phi \Psi \boldsymbol{\gamma}\|_2 \leq \varepsilon \quad (2)$$

where $\Phi \Psi$ is called the equivalent dictionary. The high-dimensional image is recovered by multiplying the dictionary with the sparsest solution: $\hat{\mathbf{x}} = \Psi \hat{\boldsymbol{\gamma}}$.

To obtain an accurate solution for Eq. (2), the equivalent dictionary is selected or designed to satisfy the RIP [22]. In essence, the RIP condition requires that the equivalent dictionary preserves meaningful information in any k -sparse vectors, thus enabling accurate reconstruction of the image $\mathbf{x}$. While confirming whether a general matrix satisfies the RIP condition is NP-hard (non-deterministic polynomial-time hard) [23], most random matrices, such as a Gaussian matrix and Bernoulli random matrix, satisfy the RIP condition [24].

B. Single-Pixel Camera

In CS, a scalar entry y_j of the measurement vector is computed as the inner product between a row of the sensing matrix ϕ_j^T and the image vector $\mathbf{x}$. By reshaping the vectors ϕ_j^T and $\mathbf{x}$ into matrices, Φ_j and $\mathbf{X}$, respectively, each entry y_j can be computed as the element-wise product between these matrices [20]:

$$y_j = \phi_j^T \mathbf{x} = \sum \Phi_j \odot \mathbf{X} \quad (3)$$

This idea is used in a single-pixel camera [25], where a scene is focused on a digital micromirror device (DMD) by a convex lens, and a photodiode collects the reflected light, as illustrated in Fig. 1. The DMD works as a binary alternative of the sensing matrix, and by repeating the measurement m times with different mirror patterns in the DMD, the original scene can be reconstructed by a CS solver using the measurement vector $\mathbf{y}$. Thus, the single-pixel camera is capable of imaging a scene with only a single photodiode.

C. Analogy to Light Curve

In previous work [20], a qualitative and quantitative comparison was performed between a single-pixel camera and light curve measurement. A schematic diagram to illustrate the comparison is shown in Fig. 2.

In the single-pixel camera, as depicted in Fig. 2a, the pixel values of the scene are projected onto a DMD mirror pattern and randomly masked. Subsequently, the masked scene is focused by a positive lens on a photodiode, which measures the total brightness.

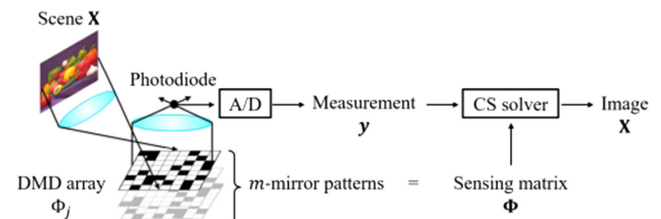


Fig. 1 Schematic of single-pixel camera (A/D, analog-to-digital converter).

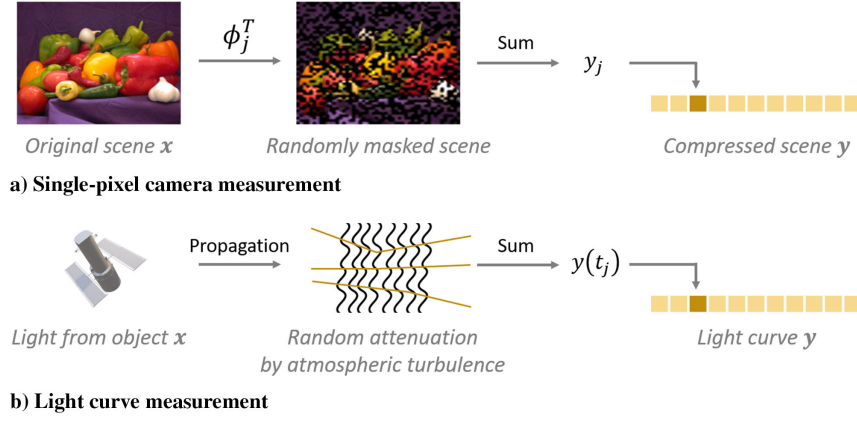


Fig. 2 Comparison of single-pixel camera and light curve measurement.

In contrast, light curve measurement, illustrated in Fig. 2b, entails the reflection of sunlight from each point on a space object, followed by the propagation of light beams through atmospheric turbulence, leading to their attenuation. This random attenuation may be considered analogous to the random masking of image entries in the single-pixel camera. These attenuated light beams are then captured by a telescope and are recorded as an intensity of the light curve, which mirrors the summation of masked image entries.

Based on this analogy, this research leverages compressed sensing to recover a resolved image of an unknown space object based on its light curve measurement. The attenuation of the light curve due to atmospheric interference is computed using PSFs computed by a rigorous wave propagation model through atmospheric turbulence. More details about the light propagation method can be found in Sec. III.

III. Light Propagation Modeling

A. Overview

The light curve measurements are simulated to fully explore the capabilities of the proposed method. This technique has the advantage of providing access to ground truth in evaluating the method. The light curve simulation method is purposely physics based to be as representative of the real observations as possible.

Specifically, this research adopts an anisoplanatic imaging simulation method to model wave propagation through free space and turbulence for a low Earth orbit (LEO) object located at 550 km. The proposed method extends the framework developed by Hardie et al. [26], which has been validated for horizontal paths of short distances (up to 7 km).

First, a spherical wave is emitted from every point on the object plane and propagated over a distance L_{free} through the atmosphere-free space. The simplicity of the medium allows the propagated wave to be analytically computed via the Fresnel diffraction integral (see [27] Chap. 6).

The optical wave is then propagated through the atmospheric turbulence from the upper atmosphere to the ground. First, the random phase fluctuations caused by the turbulence are modeled using a set of N extended phase screens, which are equally spaced over a distance L_{turb} . The extended phase screens are then cropped to create local phase screens of size $N \times N$ ($N < N_{\text{ext}}$) within a predetermined distance of the optical path from each point on the object. This two-step procedure is employed to model the spatial correlation of the local turbulence. Figure 3 shows the extended phase screens and two sets of local phase screens depicted as blue and green squares.

The propagated wave is then used to compute a PSF for each point on the object. Ideally, the PSFs are calculated for all pixels on the object plane, but the PSFs are instead calculated at sample points spaced by small pixel values on the object plane to save computational time. The PSF values at nonsample points are calculated using bilinear interpolation, which preserves the spatial correlation of the atmospheric turbulence.

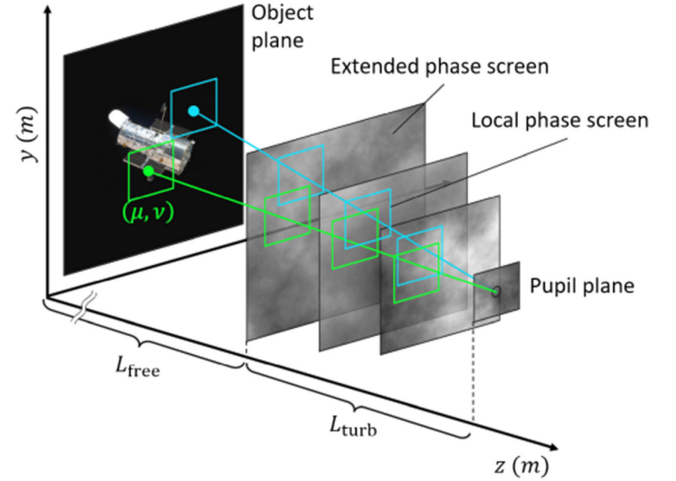


Fig. 3 Geometry of the optical wave propagation of a LEO object.

After calculating the spatially varying PSFs, the degraded image of the object is calculated as a spatially weighted average of the true object. This degraded image represents the theoretical image measured on a PSF sensor, which may or may not be the same as the ground-based telescope used for imaging. The ground-based sensor is assumed to have an aperture and resolution, resulting in a degraded, noise-affected light curve formed by an entirely unresolved image. The degraded light curve is calculated as the pixel summation of the degraded image. Therefore, the light curve measurement is a linear mapping of the true object due to the linearity of the spatially weighted average calculation. This linear light curve model is interpreted as a linearly compressed image in compressed sensing with a sensing matrix whose row represents the superposition of the PSFs.

More details are described in the following subsections. In Secs. III.B and III.C, the theories and procedures to generate the extended phase screens are explained. In Sec. III.D, a spherical wave is propagated through the phase screens to compute its complex amplitude at the pupil plane. In Sec. III.E, the method to compute PSF is detailed. In Sec. III.F, the method to compute degraded image and light curve are explained. In Sec. III.G, the degraded image model is adapted to the compressed image model in compressed sensing. Finally, in Sec. III.H, noise in the PSF map and light curve is modeled.

B. Coherence Diameter of Each Phase Screen

The extended phase screens are computed by the Fourier transform method [28] based on the implementations by Schmidt (see [27] Chap. 9).

First, the refractive index structure function $C_n^2(z)$ is chosen to characterize the atmospheric transparency over distance z from the

space object to the observer. The $C_n^2(z)$ function may be estimated by a number of methods including a numerical weather prediction simulations [29], empirical models [30], and measurement-based methods [31]. In this study, the $C_n^2(z)$ function is simplified as zero in free space over $0 \leq z < L_{\text{free}}$ and modeled as a positive constant C_n^2 over turbulence located at $L_{\text{free}} \leq z \leq L (= L_{\text{free}} + L_{\text{turb}})$ because the atmosphere contributes to refraction only up to a certain altitude [30].

Subsequently, three statistical measures of the turbulence strength (the coherence diameter r_0 , isoplanatic angle θ_0 , and log-amplitude variance σ_χ^2) are computed by integrating the $C_n^2(z)$ function multiplied by some weight functions,

$$r_0 = \left[0.423k^2 \int_0^L C_n^2(z) \left(\frac{z}{L} \right)^{5/3} dz \right]^{-3/5} \quad (4)$$

$$\theta_0 = \left[2.91k^2 L^{5/3} \int_0^L C_n^2(z) \left(1 - \frac{z}{L} \right)^{5/3} dz \right]^{-3/5} \quad (5)$$

$$\sigma_\chi^2 = 0.563k^{7/6} L^{5/6} \int_0^L C_n^2(z) \left(\frac{z}{L} \right)^{5/6} \left(1 - \frac{z}{L} \right)^{5/6} dz \quad (6)$$

where $k = 2\pi/\lambda$ is the wave number, λ is the wavelength, and L is the propagation distance. The coherence diameter r_0 is introduced by Fried [32] and is a diameter for which the turbulence starts to seriously distort the optical field. The isoplanatic angle θ_0 is an angular separation with which two point sources will have approximately

Using the r_{0_i} value, each phase screen is simulated by the Monte Carlo method based on the modified von Kármán statistics. More details are explained in the following.

The local coherence diameter z_0 , is defined to approximate Eq. (4) by its discrete counterpart,

$$\hat{r}_0 = \left[\sum_{i=1}^N \left(\frac{z_i}{L} \right)^{5/3} (r_{0_i})^{-5/3} \right]^{-3/5} \quad (7)$$

which is obtained by eliminating C_n^2 term by introducing a definition of the coherence diameter of the i th phase screen:

$$r_{0_i} = [0.423k^2 C_n^2 \Delta z_i]^{-3/5} \quad (8)$$

Solving Eq. (8) in terms of C_n^2 and substituting in the discrete versions of Eqs. (5) and (6), the isoplanatic angle and log-amplitude variance are also expressed as in terms of r_{0_i} :

$$\hat{\theta}_0 = \left[6.8794L^{5/3} \sum_{i=1}^N \left(1 - \frac{z_i}{L} \right)^{5/3} (r_{0_i})^{-5/3} \right]^{-3/5} \quad (9)$$

$$\hat{\sigma}_\chi^2 = 1.331 \left(\frac{L}{k} \right)^{5/6} \sum_{i=1}^N \left(\frac{z_i}{L} \right)^{5/6} \left(1 - \frac{z_i}{L} \right)^{5/6} (r_{0_i})^{-5/3} \quad (10)$$

Equating Eqs. (7), (9), and (10) gives a linear matrix equation as follows:

$$\underbrace{\begin{bmatrix} (\hat{r}_0)^{-5/3} \\ \frac{\hat{\sigma}_\chi^2}{1.331k^{-5/6}L^{5/6}} \\ (\hat{\theta}_0)^{-5/3} \\ \frac{\hat{\sigma}_\chi^2}{6.8794L^{5/3}} \end{bmatrix}}_{\hat{b}} = \underbrace{\begin{bmatrix} \left(\frac{z_1}{L} \right)^{5/3} & \left(\frac{z_2}{L} \right)^{5/3} & \dots & \left(\frac{z_N}{L} \right)^{5/3} \\ \left(\frac{z_1}{L} \right)^{5/6} \left(1 - \frac{z_1}{L} \right)^{5/6} & \left(\frac{z_2}{L} \right)^{5/6} \left(1 - \frac{z_2}{L} \right)^{5/6} & \dots & \left(\frac{z_N}{L} \right)^{5/6} \left(1 - \frac{z_N}{L} \right)^{5/6} \\ \left(1 - \frac{z_1}{L} \right)^{5/3} & \left(1 - \frac{z_2}{L} \right)^{5/3} & \dots & \left(1 - \frac{z_N}{L} \right)^{5/3} \end{bmatrix}}_A \underbrace{\begin{bmatrix} r_{0_1}^{-5/3} \\ \vdots \\ r_{0_N}^{-5/3} \end{bmatrix}}_s \quad (11)$$

the same PSF. Finally, the log-amplitude variance σ_χ^2 is a measure that is related to the fluctuation of optical wave amplitude.

The next step is to discretize the turbulence volume into N phase screens that are equidistant over distance L_{turb} . The position of the i th phase screen is denoted by $z = z_i$ for $i = 1, \dots, N$ as shown in Fig. 4. Each phase screen represents the turbulence-induced phase disturbance at each position. The number of phase screens is set larger than the minimum value that avoids the aliasing problem, based on the conditions in [27] (see Chap. 8). To compute the phase screens, the coherence diameters of the local turbulence r_{0_i} are computed such that, as a whole, they still satisfy the statistics in Eqs. (46).

The vector $\hat{b}$ and matrix A in Eq. (11) are already known, given the positions of phase screens and the values of $\hat{r}_0$, $\hat{\sigma}_\chi^2$, and $\hat{\theta}_0$ parameters obtained earlier. The coherence diameter of the phase screens can be obtained by solving a constrained least-squares problem,

$$\min_s \|\hat{b} - As\|_2^2 \quad (12a)$$

$$\text{s.t. } 1.331k^{-5/6}L^{5/6} \left(\frac{z_i}{L} \right)^{5/6} \left(1 - \frac{z_i}{L} \right)^{5/6} r_{0_i}^{-5/3} \leq 0.2\hat{\sigma}_\chi^2 \quad (12b)$$

$$0 \leq r_{0_i}, \quad \forall i = 1, \dots, N \quad (12c)$$

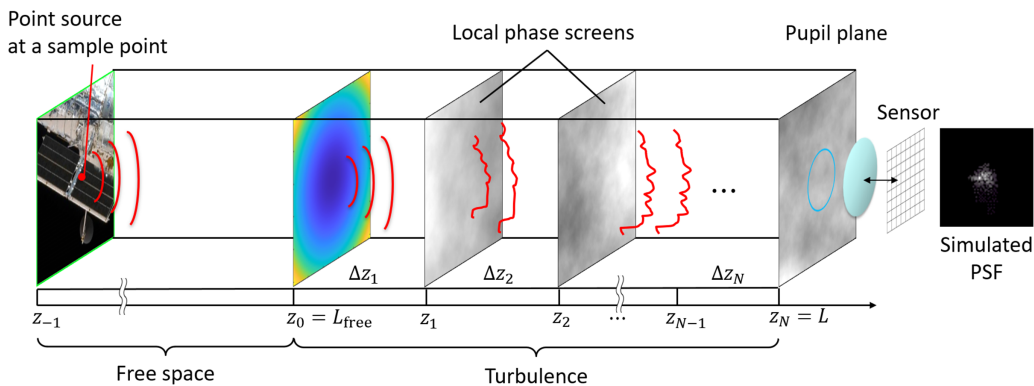


Fig. 4 Split-step propagation of a spherical wave from a sample point of an object.

where two constraints are imposed to ensure the validity of the coherence diameter values, following the Schmidt's method (see [27] Chap. 8). The first constraint is that the log-amplitude variances of the phase screens must be less than 20% of the value obtained from Eq. (6). The second constraint is that the coherence diameter of the phase screens must be nonnegative.

C. Phase Screen Realization

After computing the coherence diameters of phase screens, the extended phase screens are computed by the Monte Carlo method. The high-frequency component of the phase screen is written as a Fourier series,

$$\phi_H(x, y) = \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} c_{n,m} \exp[j2\pi(f_{x_n}x + f_{y_m}y)] \quad (13)$$

where f_{x_n} and f_{y_m} are the spatial frequencies in the x and y directions, respectively, and $c_{n,m}$ are the Fourier series coefficients. The Fourier series coefficients $c_{n,m}$ have a Gaussian distribution with a zero mean and a variance given by [27]

$$\langle |c_{n,m}|^2 \rangle = S_{\phi}^{mvK}(f_{x_n}, f_{y_m}) \Delta f_{x_n} \Delta f_{y_m} \quad (14)$$

where $\langle \cdot \rangle$ denotes the ensemble average and Δf_{x_n} and Δf_{y_m} are the grid pitch sizes in x and y directions in the frequency domain. The term S_{ϕ}^{mvK} is a modified von Kármán phase power spectral density (PSD). The PSD for the i th phase screen is given by [27]

$$S_{\phi_i}^{mvK}(\rho) = \frac{0.023e^{-\rho^2/\rho_m^2}}{r_{0_i}^{5/3}(\rho^2 + \rho_0^2)^{11/6}} \quad (15)$$

where $\rho = \sqrt{f_x^2 + f_y^2}$ is the radial spatial frequency and $\rho_m = 5.92/(2\pi\ell_0)$ and $\rho_0 = 1/L_0$ are the coefficients with respect to the outer scale L_0 and the inner scale ℓ_0 , respectively. Turbulent eddies are randomly distributed pockets of air due to the nonuniform velocity distribution. The outer and inner scales are the average sizes of the largest and smallest turbulent eddies.

The low-frequency component of the phase screen is computed by the subharmonic method [33] to improve the fidelity of the phase screens. It is generated as a sum of N_p different screens, given by

$$\phi_L(x, y) = \sum_{p=1}^{N_p} \sum_{n=-1}^1 \sum_{m=-1}^1 c_{n,m} \exp[j2\pi(f_{x_n}x + f_{y_m}y)] \quad (16)$$

The random draws of the Fourier coefficients are generated at N_p times with different grid spacings of $\Delta f_p = 1/(3^p W)$, where W is the width of the sampling grid. This sampling allows generating a screen with varieties of frequency components. The final phase screen is a sum of the high- and low-frequency components:

$$\phi(x, y) = \phi_H(x, y) + \phi_L(x, y) \quad (17)$$

D. Numerical Wave Propagation

The split-step propagation method, as depicted in Fig. 4, is used to generate a PSF that corresponds to a single sample point of the object. This method involves the use of a spherical wave and a set of $\mathcal{N}$ local phase screens, which is selected from the extended phase screens based on the geometry depicted in Fig. 3.

This study considers the wave propagation through free space over a distance L_{free} and the subsequent propagation through turbulence of a distance L_{turb} . The point source u_{pt} is modeled by a Dirac delta function, and the wave field after free-space propagation is computed using the Fresnel diffraction integral (see [27] Chap. 6):

$$\begin{aligned} u_0(x, y) &= \frac{e^{jkL_{\text{free}}}}{j\lambda L_{\text{free}}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} u_{\text{pt}}(x_{\text{pt}}, y_{\text{pt}}) e^{j\frac{k}{2L_{\text{free}}}[(x_{\text{pt}}-x)^2 + (y_{\text{pt}}-y)^2]} dx_{\text{pt}} dy_{\text{pt}} \\ &= \frac{e^{jkL_{\text{free}}}}{j\lambda L_{\text{free}}} e^{j\frac{k}{2L_{\text{free}}}(x^2 + y^2)} \end{aligned} \quad (18)$$

To model the propagation through turbulence, the wave field is propagated through a series of phase screens and free space, sequentially applying the following equation,

$$u_i(x, y; t) = [u_{i-1}(x, y; t) * h_{\Delta z_i}(x, y)] e^{j\phi_i(x, y; t)} \quad (19)$$

for $i = 1, 2, \dots, \mathcal{N}$. Here, $u_i(x, y; t)$ and $u_{i-1}(x, y; t)$ are the wave fields at the i th and $(i-1)$ th phase screens, respectively. The last term in Eq. (19) corresponds to the phase disturbance where $\phi_i(x, y; t)$ is the turbulence-induced random phase fluctuation that is modeled by the i th phase screen. It is a function of time t depending on a wind function. The operation $A * B$ represents a two-dimensional (2D) convolution of A and B . The term $h_{\Delta z_i}(x, y)$ is the impulse response of free-space propagation for Fresnel diffraction between the $(i-1)$ th and i th phase screens, which is given by

$$h_{\Delta z_i}(x, y) = \frac{e^{jk\Delta z_i}}{j\lambda \Delta z_i} e^{j\frac{k}{2\Delta z_i}(x^2 + y^2)} \quad (20)$$

Equation (19) is implemented by the angular spectrum method described by Schmidt [27], which uses the fast Fourier transform (FFT) to compute the 2D convolution. To optimize the performance of the FFT, the number of discrete sample points of phase screens N is chosen to be a power of 2. The sampling grid spacing of phase screens is determined based on the critical sampling size proposed by Voelz [34]:

$$\delta = \left(\frac{\lambda L_{\text{turb}}}{N} \right)^{1/2} \quad (21)$$

This grid spacing maximizes the use of bandwidth for Fresnel propagation simulation while avoiding the aliasing problem.

E. Point Spread Function

A PSF $h(x, y; t)$ is computed based on the complex amplitude at the pupil plane $u_{\mathcal{N}}(x, y; t)$ obtained previously. First, the complex amplitude is multiplied by an aperture mask and an exponential term that models a collimation by telescope. The product is then subjected to Fourier transform, which gives a Fourier spectrum in frequency domain (f_x, f_y) . Finally, the squared magnitude of the spectrum is computed, and it is transformed into a spatial domain by change of variables, $f_x = x/(\lambda\ell)$ and $f_y = y/(\lambda\ell)$:

$$h(x, y; t) = \left| \mathcal{F} \left\{ a(x, y) u_{\mathcal{N}}(x, y; t) e^{-j\frac{\pi(x^2 + y^2)}{\lambda L}} \right\} \right|_{f_x = \frac{x}{\lambda\ell}, f_y = \frac{y}{\lambda\ell}}^2 \quad (22)$$

Here, $\mathcal{F}$ denotes the Fourier transform operation, ℓ is the effective focal length of the PSF sensor, and $a(x, y)$ is the circular aperture mask of diameter D . This result gives a speckle image of PSF measured in extremely short exposure time (i.e., $\sim 10^1 \mu\text{s}$). The simulation is repeated for all the sample points spaced by 2 pixels, which gives a set of 65^2 speckle images of PSFs.

The obtained PSFs serve two purposes in this study. The first usage is for computing a degraded image that would be observed by a sensor at the focal plane, which will be detailed in Sec. III.F. The PSF works as a kernel that determines the blurriness of the image that would be observed with a large aperture.

The second usage is for testing the validity of the anisoplanatic imaging simulation by comparing the average of the speckle images with the analytical shape of the PSF. First, the numerical value of the average PSF is computed from the speckle images for 100 frames in long- and short-exposure cases. The key distinction between the two

exposure times is the correction of the atmospheric tilt effect [27]. If the tilt effect caused by the optical aberrations is not corrected, the resulting PSF will be representative of a long-exposure PSF. Conversely, if the tilt effect is corrected before the summation, the resulting PSF will approximate a short-exposure image. These procedures give a numerical result of average PSF for long and short exposures.

Second, the analytical result of the average PSF is computed. The Fourier spectrum of the PSF, referred to as the optical transfer function (OTF) H , is given by the product of the diffraction-limited OTF H_{dif} and the atmospheric OTF H_{atm} [35]:

$$H(\rho) = H_{\text{dif}}(\rho)H_{\text{atm}}(\rho) \quad (23)$$

$$H_{\text{dif}}(\rho) = \begin{cases} \frac{2}{\pi} \left[\cos^{-1} \left(\frac{\rho}{\rho_c} \right) - \frac{\rho}{\rho_c} \sqrt{1 - \left(\frac{\rho}{\rho_c} \right)^2} \right], & \rho \leq \rho_c \\ 0, & \text{otherwise} \end{cases} \quad (24)$$

$$H_{\text{atm}}(\rho) = \exp \left\{ -3.44 \left(\frac{\lambda \ell \rho}{r_0} \right)^{5/3} \left[1 - \alpha \left(\frac{\lambda \ell \rho}{D} \right)^{1/3} \right] \right\} \quad (25)$$

Here, $\rho_c = D/(\lambda \ell)$ is the cutoff frequency of the diffraction-limited OTF. The parameter α is a switch parameter to distinguish between long- and short-exposure cases, with $\alpha = 0$ representing a long-exposure PSF and $\alpha = 1$ representing a short-exposure PSF. Finally, the analytical result of the average PSF is computed by applying the inverse Fourier transform to the computed OTF, $h(x, y) = \mathcal{F}^{-1}\{H(\rho)\}$, and will be compared with the numerical result to validate the anisoplanatic imaging simulation.

F. Image and Light Curve

The image on the focal plane at time t in anisoplanatic imaging is denoted as $\mathbf{Z}(\mu, \nu; t)$. It is computed by a sum of the object image spatially weighted by PSFs, as opposed to a 2D convolution between a true object image and the PSF in isoplanatic imaging, which is given by

$$\mathbf{Z}(\mu, \nu; t) = \sum_k \sum_l \mathbf{X}(\mu - k, \nu - l) h_{\mu, \nu}(k, l; t) \quad (26)$$

where $\mathbf{X}(\mu, \nu)$ is the true object image on the object plane, which will be recovered later in Eq. (36); $h_{\mu, \nu}(k, l; t)$ is the space-variant PSF computed by applying Eq. (22) to a wave propagated through turbulence at time t from the point (μ, ν) of the object.

To accurately compute Eq. (26), it is necessary to adjust the grid spacing of the PSF to match the counterpart of the object model. The smallest pixel pitch size to satisfy the Nyquist–Shannon theorem is referred to as the Nyquist grid spacing. The object $\mathbf{X}$ is imaged on the PSF detector as a 2D array with a Nyquist grid spacing on the focal plane δ_f , while the PSF obtained from Eq. (22) has a grid spacing δ_{PSF} . These two grid spacings are given by

$$\delta_f = \frac{\lambda \ell}{2D}, \quad \delta_{\text{PSF}} = \frac{\lambda \ell}{\delta N} \quad (27)$$

Because these two values are different, the result of Eq. (22) is resampled to the grid spacing of δ_f before the implementation of Eq. (26).

The computed image $\mathbf{Z}$ is measured on a pixel grid of an optical sensor. In this study, the sensor grid size is assumed to be $\delta_{\text{sensor}} = 2.69 \mu\text{m}$, approximately the same size as the Nyquist pixel size on the focal plane δ_f . Thus, it is assumed that there is no information loss due to the limited resolution of the sensor. The Poisson noise inherent to the image measurement will be addressed in Sec. III.H. At this stage, it is assumed that the image has been corrected for background noise, which can be readily accomplished in practice.

The light intensity is the brightness of a nonresolved telescope image, corresponding to an image captured using a small aperture. It is computed by summing the pixel values of the image $\mathbf{Z}$,

$$y_j = \sum_{\mu} \sum_{\nu} \mathbf{Z}(\mu, \nu; t_j) \quad (28)$$

which is measured in Watts/square meters. The light curve is influenced by atmospheric degradation because of $\mathbf{Z}$ in Eq. (28). By convention [36], the brightness of the light curve is also represented as an apparent magnitude given by

$$\text{mag} = \text{mag}_{\text{sun}} - 2.5 \log_{10} \left(\frac{y_j}{I_{\text{sun}}} \right) \quad (29)$$

where $\text{mag}_{\text{sun}} = -26.74$ and $I_{\text{sun}} = 1360.8 \text{ W/m}^2$ are the magnitude and intensity of the sun at 1 astronomical unit (AU) from the Earth, respectively. A smaller magnitude corresponds to a brighter light curve.

G. Adaptation to Compressed Sensing

The degraded image model in Eq. (26) and the light intensity model in Eq. (28) are adapted for use in compressed sensing. As the spatially weighted summation in Eq. (26) is a linear operation, a flattened image model $\mathbf{z}(t_j)$ is linear in terms of a flattened object model $\mathbf{x}$, which is given by

$$\mathbf{z}(t_j) = [\mathbf{h}_1(t_j), \dots, \mathbf{h}_{65^2}(t_j)]^T \mathbf{x} \quad (30)$$

Here, $\{\mathbf{h}_l(t_j)\}_{l=1, \dots, p^2}$ is formed by shifting the centroid of each PSF sample to the corresponding source location and then flattening the resulting array. The parameter p denotes the number of PSF samples per side. The term *flattening* refers to the process of converting a 2D array into a column vector by slicing it into columns and concatenating them.

As in Eq. (28), the light intensity at time t_j is the sum of the entries in the degraded image $\mathbf{z}$. The summation of the entries can be achieved by multiplying an all-1's vector $\mathbf{1}$ with the image model:

$$y_j = \mathbf{1}^T \mathbf{z}(t_j) = \boldsymbol{\phi}_j^T \mathbf{x} \quad (31)$$

By forming a vector $\boldsymbol{\phi}_j^T$, which is a superposition of all the spatially variant PSF samples, the light intensity can be written as a linear transformation of the ideal object image $\mathbf{x}$. In this research, $\boldsymbol{\phi}_j^T$ is referred to as a PSF map, which will be used as a sensing matrix in compressed sensing.

H. Noise in PSF Map and Light Curve

This study analyzes the noise present in the PSF map and the light curve independently.

The PSF map is influenced by estimation noise. First, speckle images of guide stars in the proximity of the object are observed by telescope and imaged on the Complementary Metal-Oxide-Semiconductor (CMOS) sensor. The short-exposure time of the sensor would allow the measurements of thousands of speckle images in approximately 10^1 s. Given the speckle images, the PSF map around the object would be estimated by PSF determination techniques, such as those found in the adaptive optics community [37]. The estimation noise of the PSF map is modeled as additive white Gaussian noise with variance σ^2 . The noisy PSF map $\tilde{\boldsymbol{\phi}}_j^T$ is given by

$$\tilde{\boldsymbol{\phi}}_j^T = \boldsymbol{\phi}_j^T + \mathbf{v}^T, \quad \text{subject to } \mathbf{v}^T \sim \mathcal{N}(0, \sigma^2) \quad (32)$$

In Sec. VI, various levels of Gaussian noise are added to the PSF map to evaluate the impact on the image reconstruction.

The light curve is influenced by noise inherent to the measurement process using a charge-coupled device (CCD) or CMOS sensor. The

degraded image that accounts for the information loss and the Poisson noise is modeled as $\tilde{Z}$, which is given by

$$\tilde{Z}(\mu, \nu; t_j) = \text{Poisson}\left\{\text{uint}\left\{\frac{\pi(D/2)^2 \lambda / (\hbar c)}{g} \cdot Z(\mu, \nu; t_j)\right\}\right\} \quad (33)$$

where $\hbar$ is the Planck constant, c is the speed of light, and g is the gain of the sensor. The operator $\text{uint}\{\cdot\}$ converts the intensity of the binned image to an unsigned 32 bit integer value to model the number of bits in the sensor's analog-to-digital converter (ADC). The operator $\text{Poisson}\{\cdot\}$ generates a Poisson random variable with the value in the bracket as the mean.

The light curve affected by the finite sensor resolution and Poissonian measurement noise is then computed by summing the entries of the image $\tilde{Z}$ and then converting the unit from ADU into Watts/square meters by known physical parameters. The resulting noisy light curve model $\tilde{y}_j$ is given by

$$\tilde{y}_j = \text{adutoirr}\left\{\sum_{\mu} \sum_{\nu} \tilde{Z}(\mu, \nu; t_j)\right\} \quad (34)$$

where adutoirr is an operator for the unit conversion.

IV. Image Recovery

A. Image Recovery by TV Minimization

Using the light curve measurement and the PSF map obtained from the sensor, the proposed method reconstructs a two-dimensional image by applying the principles of compressed sensing. The light curve model discussed in Sec. III.G can be written in the form of Eq. (1) by concatenating Eq. (31) for $j = 1, \dots, m$. Each row of the sensing matrix corresponds to the PSF map at each time step t_j for $j = 1, \dots, m$. In a traditional approach, the image x can be obtained by solving Eq. (2).

There are two primary challenges associated with directly solving Eq. (2): computational cost and the design of the sensing matrix.

First, traditional reconstruction methods entail a high computational cost due to the large size of the image vector, which typically exceeds $n > 10^5$. Even for small values of m , determining $\hat{x}$ can be computationally demanding. While fast solvers exist for this problem, they often compromise reconstruction accuracy.

Another challenge lies in the design of the sensing matrix Φ . As discussed in Sec. II, the equivalent dictionary must satisfy the RIP condition to accurately reconstruct the image x . However, in LEO imaging, the PSF map, which is solely determined by atmospheric turbulence, serves as the sensing matrix. One of the popular metrics to measure the RIP condition is a mutual coherence (see [38] Chap. 2), which ranges from 0 to 1. Roughly speaking, the smaller value means the better RIP condition. While a popular sensing matrix (e.g., a Gaussian matrix) gives a mutual coherence of around 0.5, the PSF map gives 1.0. Therefore, the PSF map does not satisfy the RIP condition, resulting in inaccurate reconstruction using Eq. (2).

The first challenge is overcome by narrowing the region of interest. The location of the object is estimated using a combination of Airy disk imaging of the light curve and background correction and removal. Hereinafter, the pixel indices of the region of interest is denoted as an ordered sequence of indices $I = (i_1, \dots, i_k)$. Using this index notation, $M_I = [m_{i_1} | \dots | m_{i_k}]$ denotes a submatrix of M that consists of the columns indexed by I , in the order in which they appear in I . Similarly, a vector $v_I = (v_{i_1}, \dots, v_{i_k})^T$ denotes a sub-vector of v that includes the elements indexed by I .

The second challenge is overcome by the use of a more robust mathematical formulation, the total variation minimization. The TV is a measure of the roughness of the image [39]. It is defined as the sum of gradient of the adjacent pixel values in an image:

$$TV(X) \triangleq \sum_i \sum_j \sqrt{(X_{i+1,j} - X_{i,j})^2 + (X_{i,j+1} - X_{i,j})^2} \quad (35)$$

Equation (1) can be solved by minimizing the TV while regularizing the solution with the Dantzig selector constraint [40]:

$$\hat{X} = \arg \min_X TV(X) \quad \text{subject to} \quad \|\Phi_I^T(y - \Phi_I x_I)\|_{\infty} \leq \varepsilon \quad (36)$$

This method uses a fact that the signal to be recovered is a two-dimensional image, as opposed to Eq. (2). The TV minimization promotes the smoothness in the solution by suppressing high-frequency intensity variations. Moreover, the Dantzig selector constraint requires that the sparse representation error ($= y - \Phi_I x_I$) is not significantly correlated with any of the columns of Φ_I . This constraint encourages the error to be a white noise, leaving no traces of the columns of Φ_I . The problem in Eq. (36) is recast as a second-order cone program and solved with a log-barrier algorithm [41].

B. Image Evaluation

The recovered image is evaluated by an image quality metric called structural similarity (SSIM) index [42], which is defined by

$$SSIM(\hat{X}, X) = l(\hat{X}, X) \cdot c(\hat{X}, X) \cdot s(\hat{X}, X) \quad (37)$$

where $l(\hat{X}, X)$, $c(\hat{X}, X)$, and $s(\hat{X}, X)$ are the luminance, contrast, and structural term,

$$\begin{aligned} l(\hat{X}, X) &= \frac{2\mu_{\hat{X}}\mu_X + C_1}{\mu_{\hat{X}}^2 + \mu_X^2 + C_1}, & c(\hat{X}, X) &= \frac{2\sigma_{\hat{X}}\sigma_X + C_2}{\sigma_{\hat{X}}^2 + \sigma_X^2 + C_2}, \\ s(\hat{X}, X) &= \frac{\sigma_{\hat{X}X} + C_2/2}{\sigma_{\hat{X}}\sigma_X + C_2/2} \end{aligned} \quad (38)$$

where $\hat{X}$ is the recovered image, X is the true object image, μ is the mean, and σ is the standard deviation or covariance. C_1 and C_2 are some small constants to avoid singularity. The SSIM matches the human-perceived visual quality and is hence superior to other traditional image quality metrics, such as the mean squared error and the peak signal-to-noise ratio. The SSIM represents the image quality in the range $(-1, 1]$, where 1 denotes the perfect similarity, 0 denotes no similarity, and -1 denotes the perfect anticorrelation.

V. Simulation of Light Curve and PSF Map

A. Validation

The anisoplanatic imaging in Sec. III has been implemented to simulate a realistic light curve of the Hubble Space Telescope (HST) [43] located in low Earth orbit. The ground truth object is modeled as a static image shown in Fig. 5 by assuming that the imaging of the space object takes place so rapidly that the stabilized object's scene does not significantly change during the measurement, which can readily be accomplished by the CMOS sensor. The simulation parameters are provided in Table 1. The chosen propagation distance is a mean altitude of HST, $L = 550$ km, with $L_{\text{free}} = 500$ km and

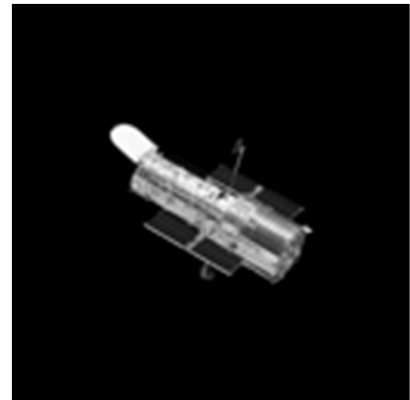
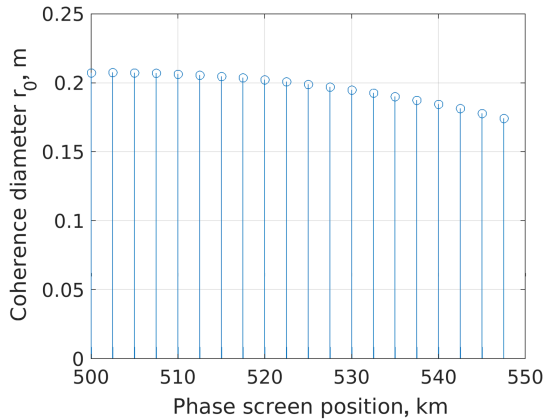


Fig. 5 True object image [43].

Table 1 Simulation parameters

Parameter	Value
Aperture diameter PSF sensor	$D = 0.61$ m
Focal length PSF sensor	$\ell = 6.25$ m
Wavelength	$\lambda = 525$ nm
Gain of sensor	$g = 12e^-/\text{ADU}$
Nyquist pixel spacing of phase screens	$\delta = 16.8$ mm
Nyquist pixel spacing of PSFs	$\delta_f = 2.69$ μm
Nyquist pixel spacing of object plane	$\delta_o = 0.237$ m
Pixel spacing of sensor	$\delta_{\text{sensor}} = 2.5$ μm
Propagation distance through free space	$L_{\text{free}} = 500$ km
Propagation distance through turbulence	$L_{\text{turb}} = 50$ km
Propagation step in turbulence	$\Delta z = 2.5$ km
Number of phase screens	$\mathcal{N} = 21$
Extended phase screen samples	$N_{\text{ext}} = 1472$ pixels
Local phase screen samples	$N = 1024$ pixels
Refractive index structure parameter	$C_n^2 = 1.0 \times 10^{-16} \text{ m}^{-2/3}$
Inner scale	$\ell_0 = 0.01$ m
Outer scale	$L_0 = 300$ m
Object model samples	129×129 pixels
Object plane side length	30.5 m
Pixel skip	2 pixels (65×65 PSF arrays)

**Fig. 6** Atmospheric coherence diameter values for phase screens.

$L_{\text{turb}} = 50$ km for free space and turbulence propagation distances, respectively.

The strength of the turbulence, or the refractive index structure parameter [44], is modeled as a step function with $C_n^2 = 1.0 \times 10^{-16} \text{ m}^{-2/3}$ over L_{turb} and nil over L_{free} . The coherence diameter, isoplanatic angle, and log-amplitude variance for the entire optical path are found to be 3.36 cm, 0.368 μrad , and 2.20.

A set of 21 extended phase screens is computed using constrained least-squares optimization. They are located equidistantly over $L_{\text{turb}} = 50$ km. The coherence diameter of each phase screen is shown in Fig. 6. The majority of the screens have coherence diameters of around 20 cm, which is significantly larger than that of the entire optical path, 3.36 cm. This discrepancy can be attributed to the influence of free-space propagation. The last extended phase screen is constrained to have $r_{0_N} = \infty$ to avoid the excessive tilt correlation [26].

Subsequently, numerical wave propagation through free space and turbulence is implemented by the angular spectrum method [27]. The spherical waves are propagated from sample points spaced by two pixels on the object plane of 129×129 pixels, resulting in 65×65 propagated spherical waves.

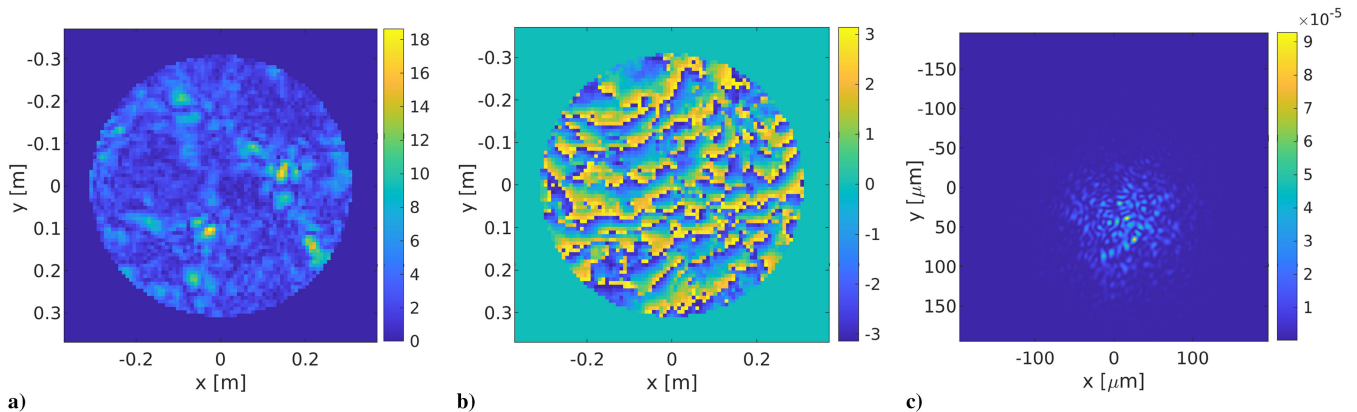
Figure 7 shows one example of the propagated wave. The irradiance and phase of the propagated wave are well sampled on the sampling grid. The smallest local phase screen sample size to avoid an aliasing problem is 1024 pixels. This large phase screen size slows down the computation of the angular spectrum method, which needs to be repeated 65^2 times in one simulation. The bottleneck of the computation, the FFT computation, is parallelized on an NVIDIA A100 Tensor Core GPU. As a result, the average computation time for obtaining an image is around 2.4 min.

Finally, the simulations are validated through a comparison between the numerical and analytical PSF, as detailed in Sec. III.E. The numerical PSF is computed by averaging 100 speckle images generated through simulations that use distinct sets of extended phase screens to ensure randomness. The analytical PSF is computed based on Eqs. (23–25). The results of this validation are presented in Fig. 8. The numerical PSF exhibits good agreement with the analytical PSF in both the long- and short-exposure cases.

B. Illustration

To illustrate the impact of object size on image resolution, a simulation has been performed using four distinct object sizes: 15, 10, 6, and 1 m for the actual Hubble Space Telescope. The simulation results are shown in Fig. 9, which shows the images observed by the PSF sensor in the absence of any Poisson noise. The top row of Fig. 9 displays the results of numerical wave propagation through free space, while the bottom row exhibits the results when atmospheric turbulence is considered. It is worth noting that the total propagation distance is fixed at $L = 300$ km for a fair comparison. As depicted, the presence of turbulence leads to significant image blurring and distortion, while the free-space propagation has minimal impact on the object's appearance. Furthermore, the shape of the object remains recognizable in Figs. 9e and 9f, but it is rendered nearly unrecognizable in Figs. 9g and 9h, where it is relegated to the nonresolved regime.

The temporal change of optical turbulence can be also simulated by shifting the extended phase screens by several pixels for 50 frames. The magnitude of the screen shift is determined based on

**Fig. 7** Example of the propagated wave and PSF: a) irradiance of the propagated wave, b) phase of the propagated wave, and c) PSF.

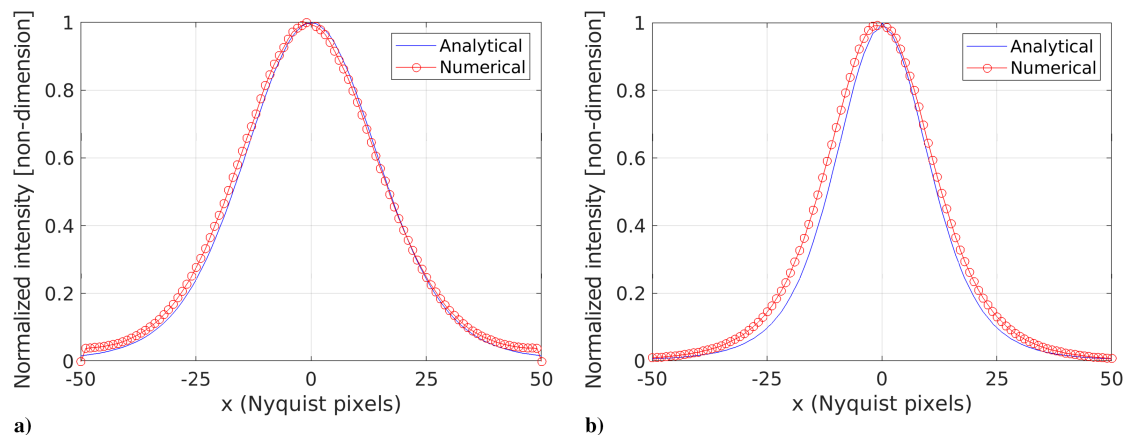


Fig. 8 Validation of average PSF for various exposure times: a) long-exposure case ($\alpha = 1$) and b) short-exposure case ($\alpha = 0$).

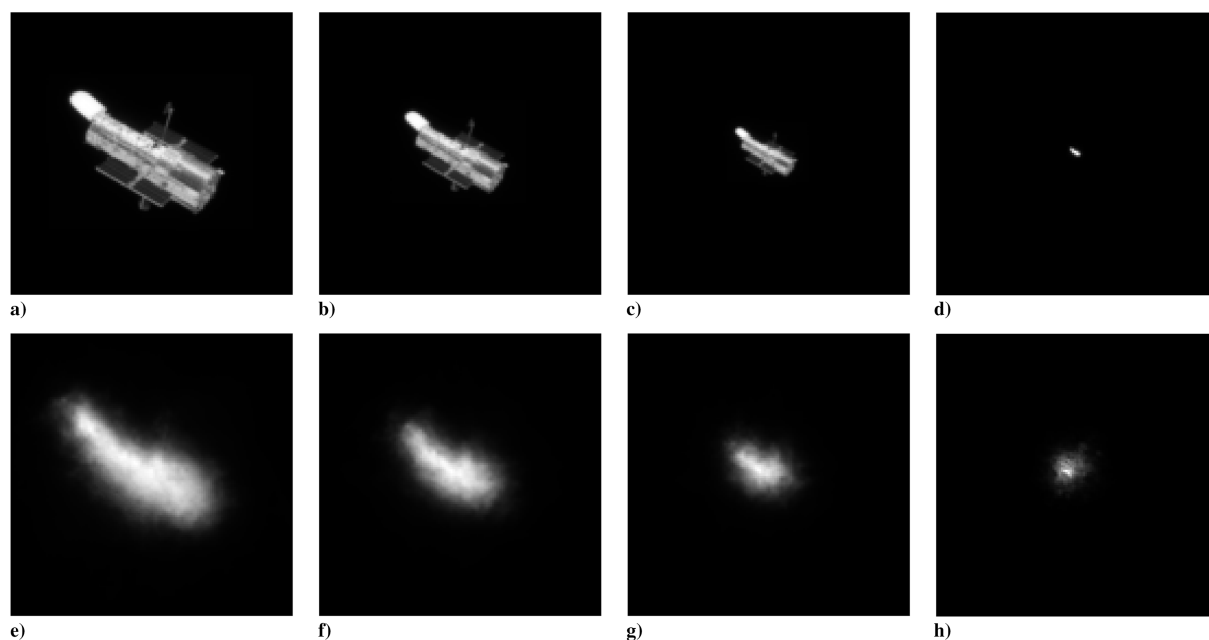


Fig. 9 Comparison of degraded images with and without atmosphere. Top row: without atmosphere; bottom row: with atmosphere. Columns vary by object size: 15, 10, 6, and 1 m from left to right.

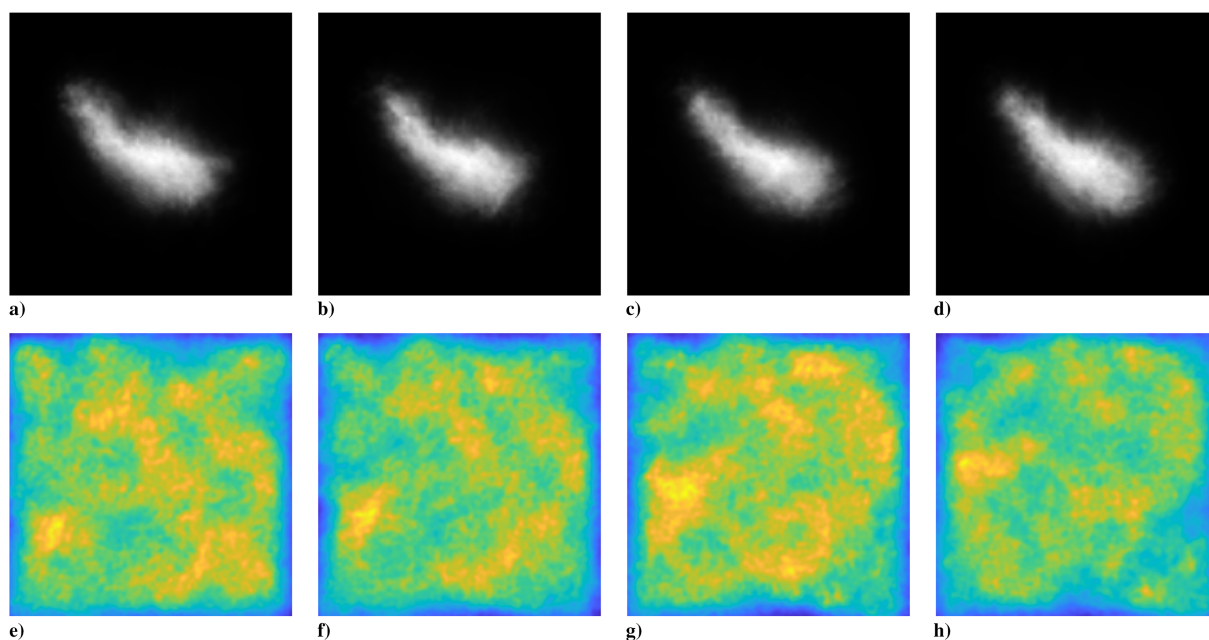


Fig. 10 Impact of slow wind speeds on the degraded image and PSF map: a–d) images at frame 1, 10, 20, and 30 and e–h) PSF map at frame 1, 10, 20, and 30.

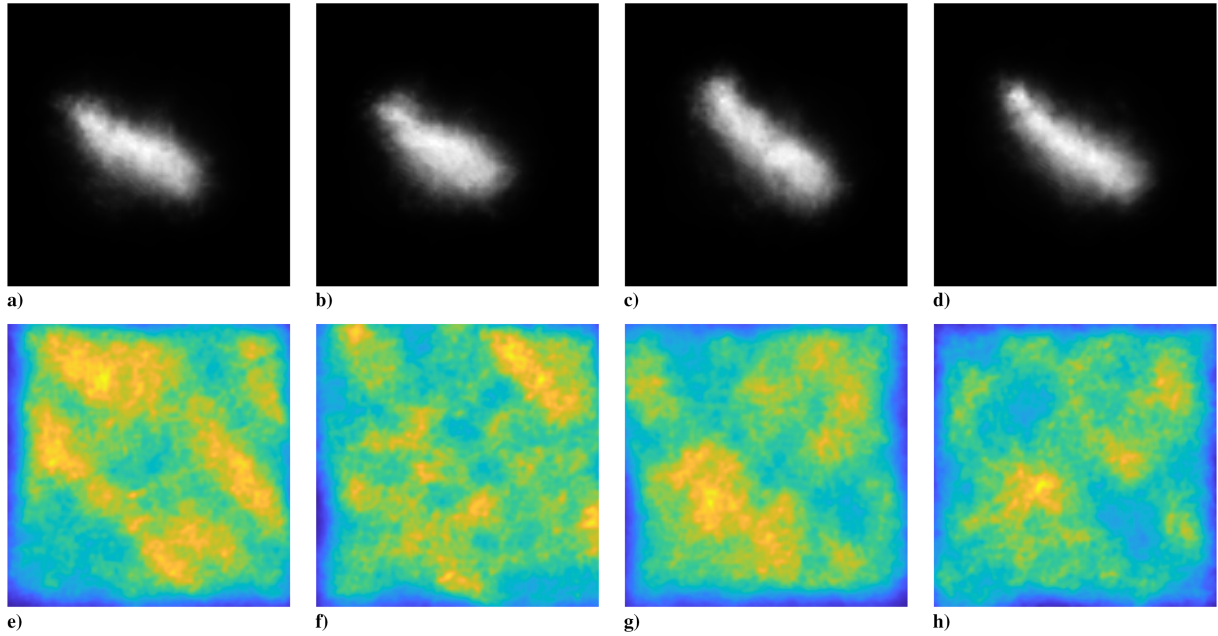


Fig. 11 Impact of high wind speed on the degraded image and PSF map: a–d) images at frame 1, 10, 20, and 30 and e–h) PSF map at frame 1, 10, 20, and 30.

the wind speed of 1 m/s and the frame rate of 40 frames/s. The object size is kept fixed at 13.2 m for this particular illustration. The results of the images and corresponding PSF maps at four distinct times are presented in Fig. 10. The wind blows from the bottom to the top of the images. The results reveal that the warping and brightness of the object are subject to random fluctuations due to the temporal change of the PSFs. Consequently, the observed light intensity also fluctuates over time. Figure 12a displays the normalized light curve obtained from this simulation, where the intensity is normalized by the light curve intensity at the initial frame. Given the slow wind speed, the light curve fluctuates almost continuously. However, in reality, wind speeds are on the order of 10 m/s, which would cause the light to pass through new sets of turbulence layers that are independent of the previous layers. This phenomenon is considered in Fig. 11, which displays the images and corresponding PSF maps for higher wind speeds. Although the differences on the PSF maps appear negligible, the corresponding light curve presented in Fig. 12b exhibits significant variations. Because the phase screens are temporally uncorrelated, the light curve shows more pronounced and discontinuous fluctuations of the intensity.

VI. Simulation of Image Recovery

A. Perfect PSF Map and Light Curve

To demonstrate the power and potential of the two-dimensional image reconstruction method developed in this paper, several realistic scenarios have been simulated. In the first scenario, the image of the HST is reconstructed solely based on its light curve and PSF maps in the absence of any noise: no Poisson noise in the light curve and perfect knowledge of the PSF map. Fast wind speeds of 20 m/s are assumed for greater realism. The light curve is assumed to be measured by a ground-based optical sensor that has the same or smaller aperture than the PSF sensor, resulting to a completely nonresolved measurement of brightness.

The simulation parameters are shown in Table 1. The object is represented in an image of size 129×129 pixels², shown in Fig. 5 as before. The HST is scaled to have an axial length of 13.2 m.

The simulation of the light curve and PSF maps is implemented in the following ways. First, a set of phase screens is generated. These phase screens are independent of the other frames to account for fast wind speeds. The nonresolved light curve is computed by summing up the pixel values of degraded image at each time frame. Using the phase screens, the degraded images of the HST and PSF maps are

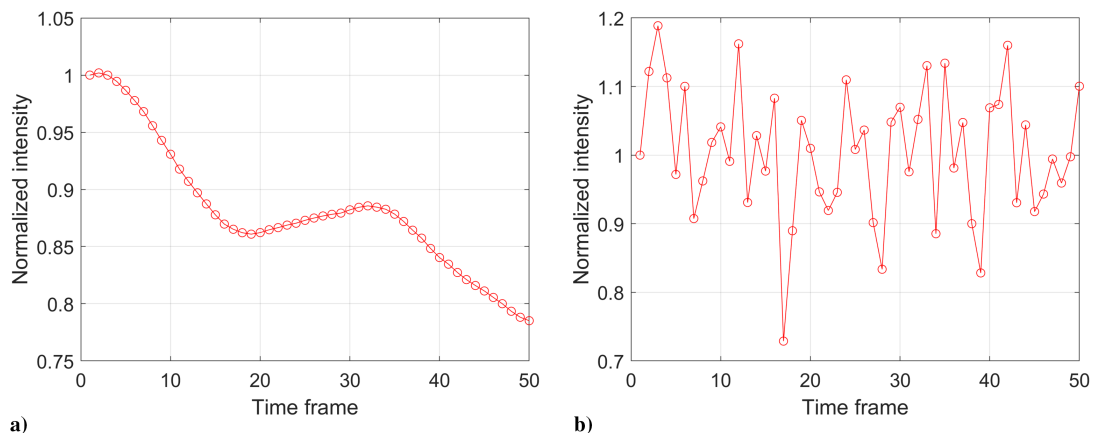


Fig. 12 Normalized light curve obtained by the anisoplanatic imaging simulation: a) results with wind speed 1 m/s and b) results with temporally uncorrelated phase screens, which corresponds to a faster wind speed.

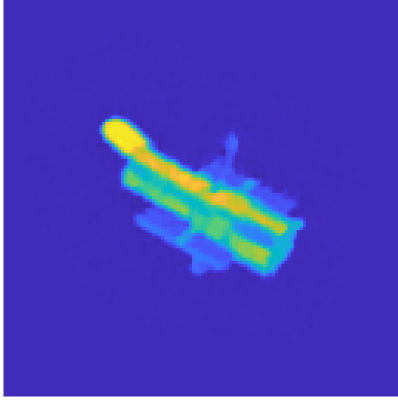


Fig. 13 Image recovery by TV minimization with perfect PSF map and light curve. SSIM = 0.971.

simulated for 2000 frames. Because the pixel spacing of the simulated PSFs δ is approximately the same as that of the focal plane δ_f , the sensor preserves the size of the PSF map to be 129×129 pixels². The PSF maps are reshaped into row vectors and concatenated to form a sensing matrix of size 2000×129^2 . The region of interest is also computed to limit the number of pixels in the recovery.

Using the light curve and the PSF maps, the image recovery simulation is conducted. The image of HST is recovered through the minimization of the total variation subject to a Dantzig selector constraint. The residual tolerance of the Dantzig selector is tuned by testing five values that are logarithmically spaced between $1e-5$ and $1e-4$ and by choosing the best value based on visual inspection. Figure 13 shows the recovered image, while the true image is shown in Fig. 5. The images are color coded for better visibility. The recovered image has the SSIM index of 0.971 and clearly shows the main body and aperture door of the HST. This result demonstrates that the PSF maps serve as a nearly perfect sensing matrix, even with only 2000 measurements, which is around 12% of the total number of unknown pixels.

B. Noisy PSF Map and Light Curve

In the following subsections, the performance of the proposed method is evaluated under more realistic conditions by reconstructing the HST image based on noisy light curve and PSF maps. The noisy PSF map is simulated by introducing white Gaussian noise to the PSF map, as described in Eq. (32). In contrast, the noisy light curve is computed using Poisson random variables to model noise inherent in photon-limited imaging, as described in Eq. (34).

Two scenarios are investigated to assess the impact of these two types of noise separately. In the first scenario, various levels of Gaussian noise are introduced to the PSF map while maintaining a constant level of Poisson noise in the light curve. In the second scenario, various levels of Poisson noise are introduced to the light curve while maintaining a constant level of Gaussian noise in the PSF map.

1. Various Gaussian Noise in PSF Map

The performance of the proposed method is evaluated under various levels of Gaussian noise in the PSF map. The noise level in the PSF map is defined in terms of the standard deviation σ in Eq. (32) in the following manner. It is assumed that the three-sigma bound of Gaussian white noise is smaller than the smallest entry of the PSF map because the entries of PSFs are positive. Therefore, the noise level is expressed as a percentage of the smallest entry, and five levels of noise have been simulated: 1, 10, 15, 20, and 25%. For each noise level, 2000 noisy PSF maps are generated and upscaled to create a sensing matrix of size 2000×129^2 . Figure 14 compares the first row of the sensing matrix with and without the additive white Gaussian noise. The added noise level is 25%, which is the highest tested.

In addition, a constant level of Poisson noise is introduced to the light curve. The image intensity is rescaled to a magnitude of

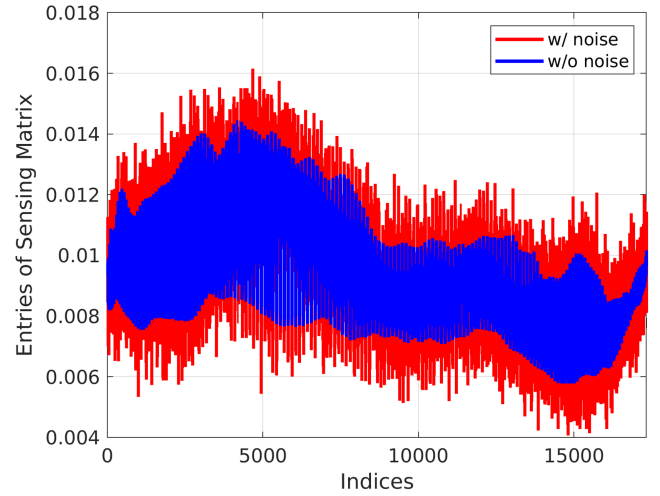


Fig. 14 Comparison of a slice of sensing matrix with (shown in red) and without (shown in blue) Gaussian noise. The noise level is 25%.

12 to represent a typical brightness of an artificial space object in LEO. The rescaled image is measured by a 32 bit ADC and converted into ADUs. A Poisson random variable is then generated based on the resulting mean value to model an image degraded by Poisson noise, as described in Eq. (33). This procedure is repeated for 2000 frames, with the light curve computed by summing entries of each noisy image.

Using the five noise levels of the PSF map and a single Poisson-degraded light curve, the image reconstruction of the HST at an altitude of 550 km is performed. The recovered HST images are shown in the top row in Fig. 15, while the true object image is displayed in Fig. 5. The image quality of the results for all the noise levels is degraded in comparison to the results reconstructed with the perfect PSF map displayed in Fig. 13. The SSIM values of the recovered images are 0.847, 0.726, 0.616, 0.550, and 0.516 when the PSF map is degraded by a Gaussian noise level of 1, 10, 15, 20, and 25%, respectively. There is a clear trend that the increased noise in the PSF map degrades the reconstruction results.

The quality of the reconstructed result is improved by applying the Block Matching and 3D filtering (BM3D) algorithm [45]. The threshold for the noise level required by the BM3D algorithm is adjusted through trial and error. The bottom row in Fig. 15 shows the images after applying the BM3D algorithm to the images directly above. The SSIM values are improved by 0.0630, 0.197, 0.310, 0.375, and 0.405 for the noise levels of 1, 10, 15, 20, and 25%, respectively. The denoising is necessary to identify the solar panels and antenna of the HST especially when the PSF map is subject to a larger noise.

2. Various Poisson Noise in Light Curve

This simulation studies the effect of different levels of Poisson noise in the light curve on the reconstruction results while keeping a constant level of Gaussian noise in the PSF map. Like in the previous simulation, the noisy PSF map is modeled using Eq. (32), but the noise level is fixed to 10%.

Four levels of Poisson noise in the light curves are modeled using Eq. (34). Initially, the image intensity of the HST is scaled to a magnitude of M . The image is subsequently degraded by the light propagation method and measured using a 32 bit ADC. A Poisson variable is then generated based on the resulting mean value. The light curve intensity at a single frame is computed by summing the entries of the noisy image. This procedure is repeated for 2000 frames for four magnitudes, $M = 12, 13, 14$, and 15 , resulting in four light curves. A larger magnitude results in a darker object and a greater impact of Poisson noise on the light curve.

Figure 16 shows the effect of Poisson noise on the object images and light curves. The columns represent relative magnitudes of 12, 13, 14, and 15 from left to right. The first row displays the random

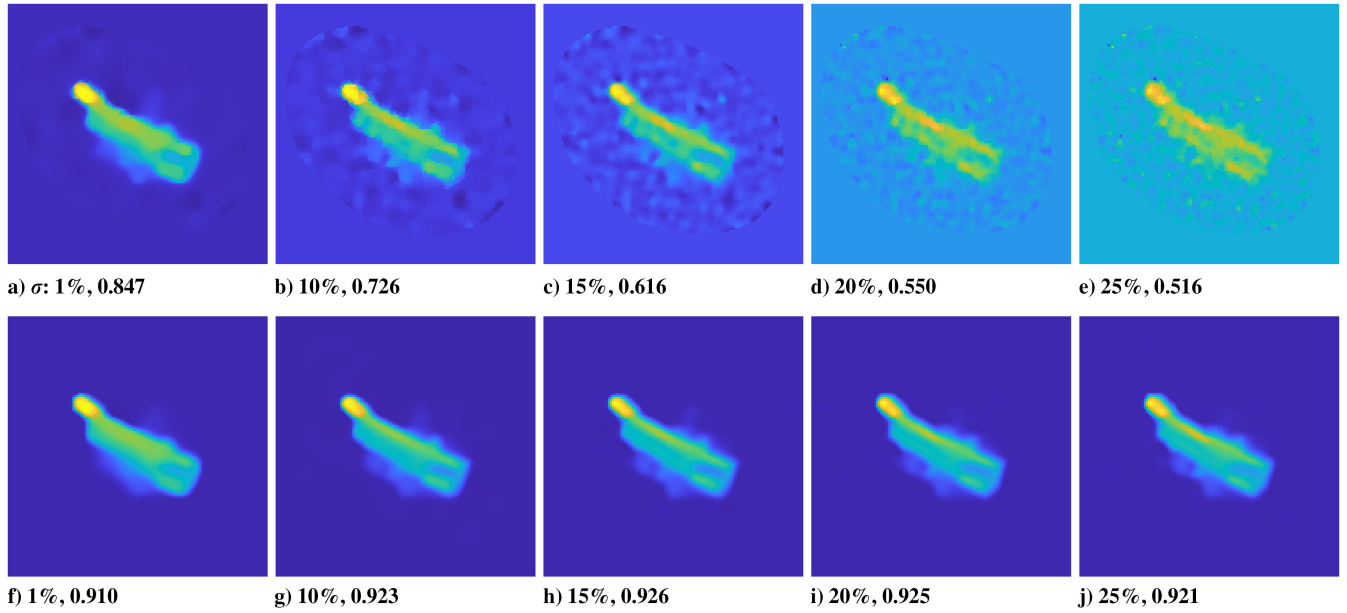


Fig. 15 Recovered (top) and denoised (bottom) images for the HST with a PSF degraded by five levels of Gaussian noise. Each caption shows the noise level and SSIM.

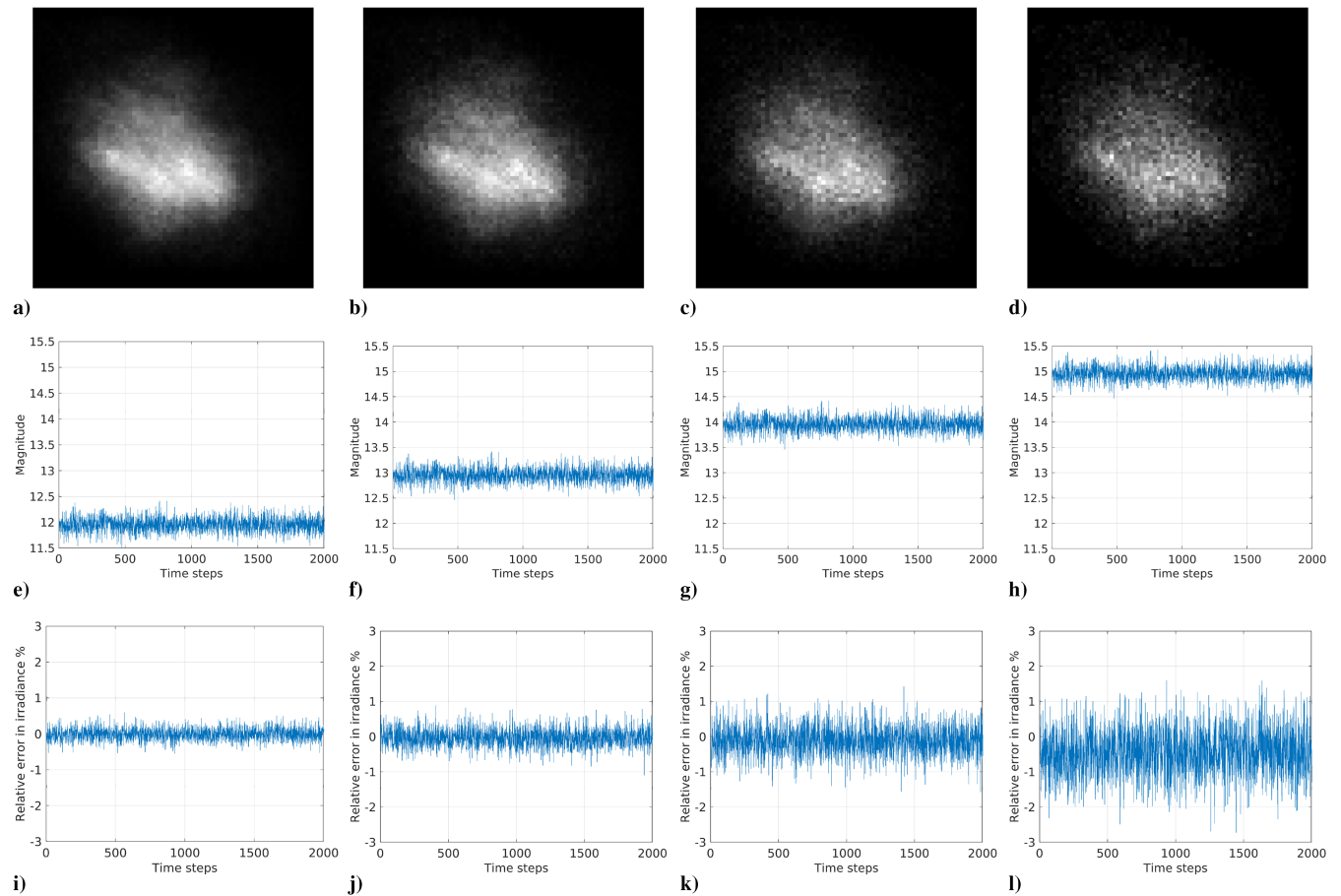


Fig. 16 Poisson noise on image measurement and its effect on light curve intensity. First row: electron counts with Poisson noise. Second row: light curve with Poisson noise. Third row: relative error of light curve. Columns vary by image brightness: magnitude of 12, 13, 14, 15 from left to right.

electron counts generated by drawing samples from Poisson distribution. The second row shows the Poisson-degraded light curve obtained by summing the electron counts and by converting it into the relative magnitude. The third row shows the relative error of the light curve in relation to the noiseless light curve. The impact of Poisson noise on the images and light curves is not evident for relative

magnitudes of 12 and 13. In these two cases, the relative error of the light curve is below 1%, as shown in Figs. 16i and 16j. However, the impact becomes more significant for objects with relative magnitudes of 14 and 15, as shown in Figs. 16k, 16l, 16k, and 16l.

Given the Poisson-degraded light curves shown in Figs. 16e–16h and a sensing matrix with a noise level of 10%, the image

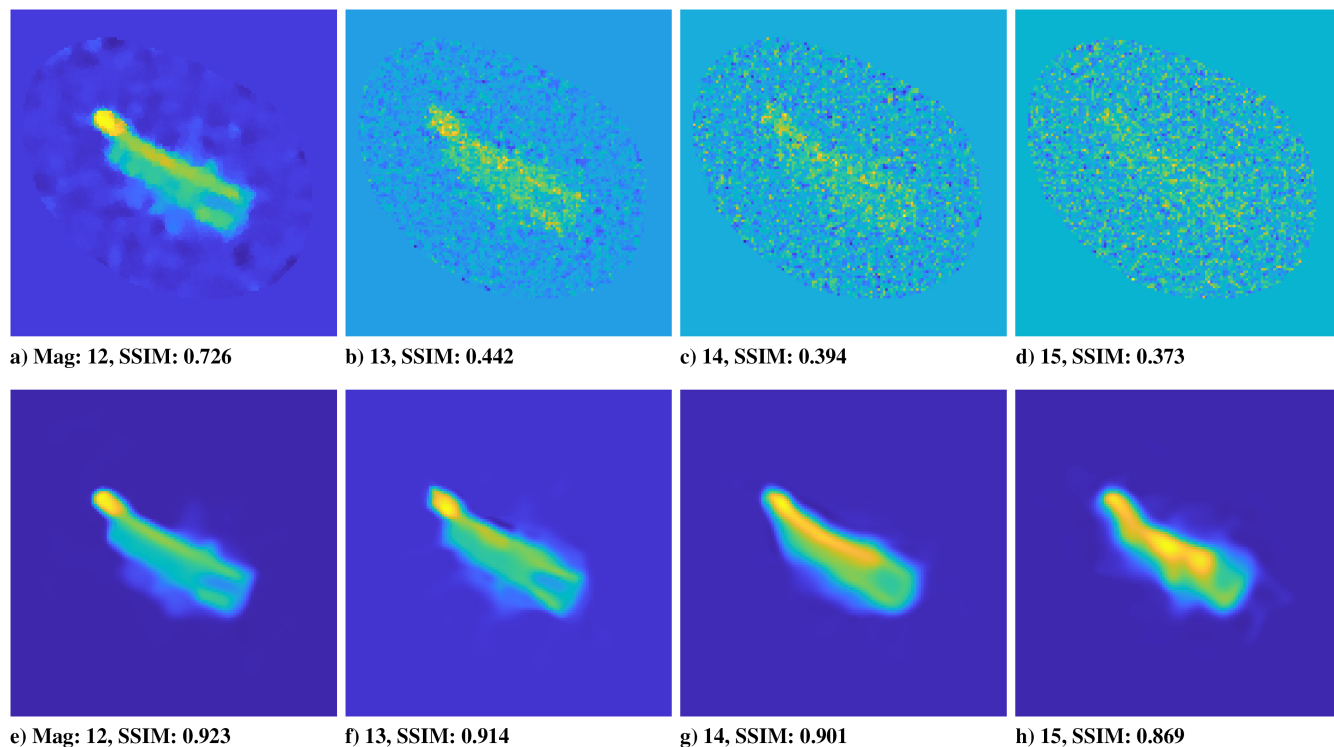


Fig. 17 Recovered (top) and denoised (bottom) images for the HST with various brightness. Each label shows the relative magnitude and SSIM.

reconstruction of the HST at an altitude of 550 km is performed. The results are shown in Fig. 17. The images in the top row are reconstruction results, and the ones in the bottom row are the images after applying the BM3D algorithm. There is a clear trend that the darker objects result in lower image quality in the reconstruction results. For objects with a relative magnitude of 12, the recovered result has the SSIM of 0.726, and the details of the HST are recognizable even without denoising.

For darker cases, the reconstructed images become noisier, and the details are hard to recognize by human eyes, but they still have the similar structure as the true HST image. As in Eq. (38), the SSIM is a product of the luminance, contrast, and structural terms. The SSIM of Fig. 17d has a high structural term of 0.827 but low luminance and contrast terms of 0.522 and 0.406. Therefore, denoising is needed to improve the luminance and contrast quality. The BM3D algorithm can reveal the solar panels for the relative magnitude of 13 with a high SSIM of 0.914. For the relative magnitudes of 14 and 15, the HST details are lost after denoising, but the contrast is restored well, resulting in a high value of SSIM. For example, the contrast term of the SSIM is 0.947 in Fig. 17h.

Several studies [46,47] have highlighted the negative impact of Poisson noise in compressed sensing. The TV minimization method with the Dantzig selector constraint assumes that the noise in the light curve is linearly mapped to the measurement residual. However, because the Poisson noise is nonadditive and signal dependent, this method may not be ideal. Despite this, given the low brightness conditions considered in this study, the proposed algorithm demonstrates sufficient robustness for practical applications.

VII. Conclusions

This paper presents a method for characterizing a stabilized object in LEO by reconstructing a resolved two-dimensional image at a frozen attitude. The inputs for the method are a PSF map and an entirely unresolved light curve. The PSF map is a superposition of spatially varying PSFs of the atmospheric turbulence, which is assumed to be measurable using PSF determination techniques.

Traditional light curve inversion methods are hindered by high dimensionality and uncertainties in the solution. The proposed method adapts the compressed sensing paradigm commonly

employed in image compression. This method overcomes these challenges by recovering a two-dimensional projection of the object of interest rather than reconstructing a three-dimensional shape with additional dimensionality encompassing area albedo.

Light curves and PSFs are simulated using light propagation through free space and atmospheric turbulence to provide the best realism. Optical perturbation due to atmospheric turbulence is modeled by Monte Carlo phase screens. Illustrations of the resulting degraded imaging for various object sizes are presented.

An image of a space object in LEO is reconstructed based on the simulated light curve and PSFs without any a priori information. The incoherent PSFs obtained from the simulations are superposed to generate a PSF map, which serves as the sensing matrix. However, the sensing matrix does not fulfill the Restricted Isometry Property required for accurate signal recovery in compressed sensing. This problem is alleviated by employing the TV minimization with the Dantzig selector constraint for image recovery. Given the PSF map and the light curve without noise, nearly perfect image recovery is achieved.

The image recovery is also considered for noisy PSF maps and light curves in two cases. The first scenario involves the presence of five levels of Gaussian noise in the PSF map, with noise levels up to 25% in standard deviation, while assuming a constant level of Poisson noise in the light curve with a magnitude of 12. The larger Gaussian noise in the PSF map degrades the quality of the results. However, after denoising the image by BM3D algorithm, the details of the object becomes discernible. The second scenario involves four levels of object brightness ranging from 12 to 15 in relative magnitude, resulting in various levels of Poisson noise in the light curve while maintaining a noise level of 10% in the PSF map. Because the Poisson noise has a greater impact on the light curve for a darker object, the larger magnitude leads to a lower quality of the recovered image. However, based on the results of these two scenarios, the proposed method is robust to noise in both the light curve measurement and PSF estimation.

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